ARENA MASTER RESEARCH BLUEPRINT

Joint Hermitian Hull-Dimension Enumeration and Quantum Codes over Finite Non-Chain Rings

Purpose: A phase-gated, proof-first workflow for auditing the uploaded paper, verifying related literature, developing candidate theorems, checking computations, and preparing a defensible research manuscript.

Primary objective	New theorems with rigorous proofs
Secondary objective	Exact hull-dimension enumeration and quantum-code applications
Working title	Joint Hermitian Hull-Dimension Enumeration and Quantum-Code Construction for Constacyclic Cod
Source paper	Yadav et al. (2024), Computational and Applied Mathematics, DOI: 10.1007/s40314-024-02789-1

Important: This document is a research plan, not a claim that the proposed enumerator or its quantum-code consequences are new. Novelty must be established by literature comparison and complete mathematical proofs.

How to Use This Blueprint

Paste the blueprint into Arena and ask it to execute one phase at a time. Do not ask Arena to write the complete paper immediately. The first task is to inspect the workspace and audit the exact uploaded PDF. Every claimed file, proof, computation, compilation, or Git operation must be verified.

Execution rule: Stop at any failed or blocked gate. If the research gap is not supported or a theorem is already known, revise the direction rather than exaggerating novelty.

Project Overview

Project type: Original mathematical research. Primary objective: rigorous theorems and proofs. Secondary objective: exact hull-dimension enumeration and valid quantum-code applications.

Source paper: “Hermitian hull of constacyclic codes over a class of non-chain rings and new quantum codes,” Shikha Yadav, Ashutosh Singh, Habibul Islam, Om Prakash, and Patrick Solé, Computational and Applied Mathematics (2024), DOI 10.1007/s40314-024-02789-1.

0. Non-Negotiable Research Rules

- Read the complete source paper, including definitions, theorem statements, hypotheses, proofs, examples, tables, references, and conclusions.
- Never claim a gap just because a result is absent from the source paper; compare the wider literature.
- Never invent a theorem, citation, DOI, parameter, program output, or verification result.
- Separate source-paper results, published results, standard consequences, proposed conjectures, newly proved results, and computational observations.
- Do not present a restatement, reparameterization, special case, or routine corollary as a new theorem.
- Every theorem must have precise hypotheses, a complete proof, boundary-case analysis, and comparison with prior work.
- Never assume a Gray map preserves distance, Hermitian hull dimension, or all duality properties without proof.
- A hull-dimension distribution does not automatically determine minimum distance.
- If files, tools, or sources are unavailable, report the limitation instead of filling gaps with guesses.
- Do not overwrite original files or push changes to a public repository without authorization.

Phase 0 — Workspace Inspection and Source Integrity

Inspect the available workspace, uploaded PDF, project directories, Git repository, and existing artifacts. Preserve original files and identify tool availability before starting.

- Record the exact source-paper filename and, if possible, SHA-256 hash.
- Inventory prior manuscripts, blueprints, audit reports, code, and conflicting versions.
- Check whether Python, Magma, finite-field libraries, LaTeX, bibliography tools, and PDF utilities are actually available.
- Create a clean project directory if needed; never infer that a file exists from a planned filename.

Deliverables: PROJECT_PLAN.md; SOURCE_INVENTORY.md; SOURCE_PAPER_METADATA.md; ENVIRONMENT_CAPABILITIES.md; PHASE_STATUS.md.

Gate 0: The exact source PDF and the usable research environment are identified. If the PDF cannot be accessed, stop and request it.

Phase 1 — Complete Source-Paper Mathematical Audit

The source paper studies Hermitian hulls of lambda-constacyclic codes over a class of finite non-chain rings, including possible hull dimensions, Hermitian LCD conditions, dual-containing conditions, a matrix-based Gray map, and quantum-code constructions. Its ring is written as

$$R_e = F_{q^2} + uF_{q^2} + \ldots + u^{e-1}F_{q^2},\ u^e = 1,$$

under the paper's exact assumptions, including e | (q+1) and specified component multipliers. Do not generalize those assumptions without proof.

1A. Extract all relevant results

- Record theorem/lemma/proposition/corollary number and exact mathematical meaning.
- Record all hypotheses, notation, proof method, consequences, and restrictions.
- Classify whether each result concerns possible dimensions, exact counts, LCD properties, dual-containing properties, or quantum codes.
- Record how each result may be used in the proposed project.

Deliverables: SOURCE_THEOREM_REGISTER.md; SOURCE_MATHEMATICAL_AUDIT.md; SOURCE_NOTATION_GLOSSARY.md; SOURCE_EXAMPLE_RECALCULATION.md.

1B. Mathematical checks

- Verify ring decomposition and assumptions.
- Check generator-polynomial factorization, multiplicities, and conjugate-reciprocal operation.
- Check hull-dimension formulas and Hermitian LCD criteria.
- Check dual-containing conditions and Gray-map properties.
- Recalculate representative examples and identify unclear or unverified arguments.

Gate 1: Central source-paper results and the assumptions needed for the new work are understood and documented.

Phase 2 — Related Literature and Novelty Audit

Search scholarly sources, publisher pages, DOI records, full texts, cited references, and forward citations where accessible. Search the literature on Hermitian hulls, exact hull enumeration, finite non-chain rings, idempotent decompositions, conjugate-reciprocal factor orbits, LCD and dual-containing codes, Gray maps, quantum constructions, Galois hulls, and repeated-root constacyclic codes.

2A. Literature comparison matrix

Create a table with columns: Paper; Ring or field; Code family; Hull type; Main theorem; Exact enumeration?; Quantum construction; Difference from proposed work.

2B. Novelty tests

- Check whether the same constituent decomposition or product formula already exists.
- Check whether the same local factor-orbit generating functions or total distribution are known.
- Check whether dual-containing enumeration and the same quantum consequences have already been derived.
- Read complete theorem statements where possible; abstracts and search snippets alone do not establish absence of a result.
- Do not treat the phrase “joint enumerator” as a novelty argument.

Classification: N1 = apparently new with documented comparison; N2 = potentially new but needs further checking; K = known or routine; F = false under proposed assumptions; U = unresolved.

Deliverables: LITERATURE_REVIEW.md; LITERATURE_COMPARISON_MATRIX.csv; NOVELTY_ASSESSMENT.md; REFERENCE_VERIFICATION_REPORT.md.

Gate 2: The proposed main contribution is distinguishable from the closest known results. Otherwise revise the research question.

Phase 3 — Precise Mathematical Formulation

Let $C_\ell(n, \alpha_\ell)$ denote the admissible α_ℓ-constacyclic codes of length n over F_{q^2}, under the source paper's actual assumptions. For a constituent code D_ℓ, define

$$\mathrm{Hull}_H(D_\ell) = D_\ell \cap D_\ell^{\perp_H},\ h_\ell(D_\ell) = \dim_{F_{q^2}} \mathrm{Hull}_H(D_\ell).$$

Investigate the proposed constituent enumerator

$$H_\ell(z_\ell) = \Sigma_{D\ell \in C\ell(n,\alpha\ell)}\, z_\ell^{h\ell(D\ell)},$$

and the proposed joint enumerator

$$H_{\mathrm{joint}}(z_1,\ldots,z_e) = \Sigma_{\text{admissible } C} \prod_{\ell=1}^{e} z_\ell^{h\ell(D\ell)}.$$

These are proposed definitions, not claims that the source paper already established such enumerators.

3A. Establish well-definedness

- Define precisely what counts as a distinct code.
- Prove uniqueness of constituent representation where required.
- Prove whether constituent choices are independent.
- Prove how the Hermitian dual and hull behave under decomposition.
- Specify the field over which every dimension is taken.
- Distinguish counting codes from counting generator-polynomial descriptions or equivalence classes.

Investigate $H_{\mathrm{total}}(z) = H_{\mathrm{joint}}(z,\ldots,z) = \Sigma_j N_j z^j$, where N_j counts distinct admissible ring codes with total hull dimension j. Prove the relationship between total and constituent dimensions before using this interpretation.

Deliverable: MATHEMATICAL_FRAMEWORK.md.

Phase 4 — Factor-Orbit Enumeration

Begin with a tractable square-free setting satisfying gcd(n,p)=1, where p is the field characteristic, together with all other hypotheses required by the chosen constacyclic multiplier. Do not assume this case covers repeated-root lengths.

- Factor $x^n - \alpha\ell$ over F_(q²) for each constituent multiplier.
- Determine the action of the conjugate-reciprocal operation on irreducible factors.
- Classify fixed factors and paired orbits.
- List admissible generator-polynomial exponent choices.
- Derive the hull contribution of each choice from a proved formula.
- Define local orbit polynomials P_O(z) = Σ_a z^(h_O(a)).
- Prove local choices are independent before multiplying their generating polynomials.
- Derive explicit Hℓ(z) and compare it with existing formulas.
- Study repeated-root cases only after proving and verifying the square-free case.

Deliverables: FACTOR_ORBIT_ANALYSIS.md; LOCAL_ENUMERATOR_DERIVATION.md; CONSTITUENT_ENUMERATOR_FORMULA.md; REPEATED_ROOT_EXTENSION.md only if actually developed.

Phase 5 — Main Theorems and Complete Proofs

Candidate Theorem A: Componentwise factorization

Prove a factorization of the joint enumerator into constituent enumerators only if the code-level bijection, dual decomposition, hull behavior, and independence assumptions justify it. Assess whether the result is new or routine.

Candidate Theorem B: Exact hull multiplicity formula

Derive explicit coefficients N_i from factor-orbit data and admissible generator-polynomial exponents. Prove completeness, absence of double counting, boundary cases, and all dependencies on n, q, p, e, and multipliers.

Candidate Theorem C: Dual-containing restricted enumeration

Investigate a restricted generating function counting codes satisfying $C^{\perp H} \subseteq C$. Define image dimension and distance precisely. If exact distance enumeration is infeasible, do not fabricate a closed form; begin with a dimension-only restricted enumerator.

Candidate Theorem D: Quantum-code consequence

For a Hermitian dual-containing linear code over F_{q^2} with parameters $[N,K,d]_{q^2}$, the standard Hermitian construction gives parameters of the form

$$[[N, 2K - N, \geq d]]_q,$$

under the construction's hypotheses. Verify the exact field convention, Gray-image length, dimension, distance bound, and dual-containing condition. A hull enumerator alone does not determine minimum distance.

Required proof standard

- Exact statement, definitions, hypotheses, and supporting lemmas.
- Complete proof with all cases and boundary conditions.
- Explicit identification of source-paper results used.
- Checks of fields, dimensions, polynomial divisibility, and duality.
- Comparison with the closest known theorem.
- Counterexample search for weakened assumptions.

Deliverable: MAIN_THEOREMS_AND_PROOFS.md. Symbolic calculations or finite searches do not replace proof.

Phase 6 — Dual-Containing Enumeration and Quantum Application

Keep separate: (i) counting codes by hull dimension; (ii) identifying Hermitian dual-containing codes; and (iii) determining which Gray images produce useful quantum parameters.

- Derive factor-exponent conditions for C^(⊥H) ⊆ C.
- Record ring and field parameters, length, generator polynomials, dimensions, Gray-map definition, image length/dimension, distance or verified lower bound, dual-containing verification, quantum parameters, Singleton bound/defect where relevant, and known-code comparison.
- Do not infer improved minimum distance from hull dimension alone.
- Do not confuse a quantum code over Fq with a classical code over F_(q²).

Deliverables: DUAL_CONTAINING_ENUMERATION.md; QUANTUM_CONSTRUCTION_THEOREM.md; QUANTUM_CODE_PARAMETER_TABLE.csv; QUANTUM_PARAMETER_VERIFICATION.md.

Phase 7 — Independent Computational Verification

Use Magma if it is genuinely available and Python for independent enumeration and exact combinatorial calculations where appropriate. If Magma is unavailable, state this explicitly.

- Build a direct enumeration path that constructs codes and computes hull intersections independently of the proposed formula.
- For several small admissible parameter instances, factor polynomials, enumerate generator polynomials, compute Hermitian duals/hulls, and compare observed counts with predicted coefficients.

- Include zero and nonzero hull dimensions, differing orbit structures, dual-containing cases where they exist, boundary cases, and negative tests.
- Recheck Gray images, image parameters, and quantum claims.
- Record software versions, exact commands, input files, raw outputs, runtimes, and limitations.
- Finite tests support the proof but do not establish universal correctness.

Deliverables: code/; data/; COMPUTATIONAL_VERIFICATION.md; REPRODUCIBILITY.md.

Phase 8 — Research-Gap and Contribution Decision

Evaluate each candidate contribution: explicit constituent generating polynomial; exact total distribution; joint constituent distribution; dual-containing enumeration; quantum existence/counting theorem; repeated-root generalization.

For every contribution record its exact claim, closest known result, precise difference, proof status, computational status, remaining assumptions, novelty evidence, and whether it belongs in the main paper. If all candidates are known or routine, stop and propose a better-supported direction.

Deliverable: RESEARCH_GAP_AND_CONTRIBUTION_DECISION.md.

Phase 9 — Final Manuscript

Write the complete manuscript only after the core results pass the gates. Suggested sections:

- 1. Introduction: motivation, related work, precise gap, and contributions.
- 2. Algebraic Preliminaries: ring, constacyclic codes, Hermitian duals/hulls, generator polynomials, Gray map.
- 3. Constituent Hull-Dimension Analysis: factorization and orbit contributions.
- 4. Joint Hull-Dimension Enumeration: definitions, theorems, proofs, special cases.
- 5. Dual-Containing Enumeration and Quantum Codes.
- 6. Computational Algorithm and Verification.
- 7. Examples and Discussion: reproducible results and comparisons.
- 8. Conclusion and Future Work: only supported claims.

Use precise academic English, consistent notation, verified citations, complete proofs, and an abstract matching the established results. Avoid unsupported words such as “first,” “optimal,” “best possible,” and “unprecedented.”

Deliverables: paper/main.tex; paper/references.bib; paper/main.pdf only if compilation succeeds; paper/README.md. If no PDF compiler is available, report the limitation and provide the LaTeX source package.

Phase 10 — Final Independent Audit

- Check every theorem, hypothesis, proof, factor orbit, polynomial identity, dimension, and dual-containing criterion.
- Verify Gray-map claims and quantum parameters under the exact construction.
- Confirm computational results match the manuscript tables.
- Compare every main theorem with its closest prior result.
- Verify reference metadata and ensure cited claims are supported.
- Re-run principal tests and record commands and versions.

Deliverables: FINAL_MATHEMATICAL_AUDIT.md; FINAL_NOVELTY_AUDIT.md; FINAL_REFERENCE_AUDIT.md; FINAL_REPRODUCIBILITY_AUDIT.md; FINAL_CHANGELOG.md; PACKAGE_MANIFEST.md.

Phase 11 — Git and Project Delivery

- Inspect current branch and working-tree status; preserve unrelated changes.

- Use a dedicated project directory or branch if appropriate.
- Do not delete prior artifacts or overwrite unrelated files.
- Do not push, publish, or create a public release without explicit authorization.
- Do not place credentials or secrets in the repository.
- Record commit hashes only for commits actually created.
- If a ZIP is created, list its actual contents and verify that it opens.

Suggested structure: research_project/README.md; project plan and phase status; source_audit/; literature/; theory/; proofs/; code/; data/; verification/; paper/; logs/; package manifest.

Execution Protocol and Stop Conditions

- Use statuses PASS, FAIL, BLOCKED, and PROVISIONAL, with evidence for every status.
- At the end of each phase report inspected materials, established facts, actual files created/changed, evidence, unresolved issues, status, and next step.
- Do not mark PASS because a document was written.
- Do not start the full manuscript until source audit, literature comparison, at least one substantive proved theorem, defensible novelty, and independent small-case verification are complete.
- If these conditions fail, deliver a feasibility report rather than a misleading manuscript.

FIRST ACTION FOR ARENA

Execute Phase 0 and Phase 1 only. Inspect the workspace, locate and read the exact uploaded PDF, extract the complete theorem register, and create the source audit. Do not draft the new paper or claim novelty. Verify every deliverable exists, report exact paths and unresolved issues, and stop after reporting the Phase 1 status.

Short Follow-Up Prompt for Arena

Execute Phase 0 and Phase 1 only from the master blueprint. Inspect the actual uploaded PDF and workspace, read the full source paper, and create the source inventory, theorem register, notation glossary, and mathematical audit. Do not draft the new paper or claim novelty yet. Verify every deliverable exists and report exact paths, unresolved issues, and the evidence supporting each conclusion. Stop after reporting the Phase 1 status.